परीक्षेचे नांव : सहायक प्राध्यापक, रसायनशास्त्र, अस्वायत्त शासकीय अभियांत्रिकी महाविद्यालय, महाराष्ट्र अभियांत्रिकी महाविद्यालयीन शिक्षक सेवा, गट -अ, चाळणी परीक्षा-2014

परीक्षेचा दिनांक : 27 जुलै, 2014

महाराष्ट्र लोकसेवा आयोगामार्फत सहायक प्राध्यापक, रसायनशास्त्र, अस्वायत्त शासकीय अभियांत्रिकी महाविद्यालय, महाराष्ट्र अभियांत्रिकी महाविद्यालयीन शिक्षक सेवा, गट -अ, चाळणी परीक्षा-२०१४ या चाळणी परीक्षेच्या प्रश्नपत्रिकेची उत्तरतालिका उमेदवारांच्या माहितीसाठी संकेतस्थळावर प्रसिध्द करण्यात आली होती. उमेदवारांनी अधिप्रमाणित (Authentic) स्पष्टीकरण / संदर्भ देऊन पाठविलेली लेखी निवेदने, तसेच तज्ज्ञांचे अभिप्राय विचारात घेऊन आयोगाने उत्तरतालिका सुधारित केली आहे. या उत्तरतालिकेतील उत्तरे अंतिम समजण्यात येतील. यासंदर्भात आलेली निवेदने विचारात घेतली जाणार नाहीत व त्याबाबत कोणताही पत्रव्यवहार केला जाणार नाही, याची कृपया नोंद घ्यावी.

उत्तरतालिका - KEY

Question Paper Preview

Notations:

1. Options shown in green color and with ✔ icon are correct.
2. Options shown in red color and with ✖ icon are incorrect.

- Question Paper Name: Assistant Professor Chemistry ACTUAL final
- Creation Date: 2014-07-23 14:07:02
- Cut Off: 10
- Duration: 60
- Number of Questions: 100
- Status: Sealed

Group 1

Number of optional sections to be attempted: 0, Group Maximum duration : 0, Group Minimum duration : 60, Revisit allowed for view? : No, Revisit allowed for edit? : No, Break time: 0

Assistant Professor Chemistry

Section type : Online, Number of Questions to be attempted:100, Mandatory or Optional: Mandatory

Subsection : 1, Question Shuffling Allowed : No

Q.1 **Question id : 8513 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)**

The correct range of wavelengths of visible region in the electromagnetic spectrum is :

Options :

1. ✖ 100 – 200 nm
2. ✖ 100 – 400 nm
3. ✔ 380 – 780 nm
4. ✖ 480 – 580 nm

Q.2 **Question id : 8514 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)**

According to Hooke's law, the stretching vibrational frequency ($\bar{\nu}$) of a bond is represented by which of the following equation?

Options :

1. ✔ $\bar{\nu} = \frac{1}{2\pi c}\sqrt{K/\mu}$

2. ✖ $\bar{\nu} = \frac{1}{4\pi C}\sqrt{K/\mu}$

3. ✖ $\bar{\nu} = 2\pi C/\sqrt{K\mu}$

4. ✖ $\bar{\nu} = 2\pi C/K\mu$

Q.3 Question id : 8515 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

NMR spectra of which of the following compound will give two proton signals?

Options :

1. ✔ 2 – Chloropropane
2. ✖ 1 – Chloropropane
3. ✖ 2 – Bromopropene
4. ✖ Vinyl chloride

Q.4 Question id : 8516 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Which of the following spectroscopy technique is used to elucidate structure especially of conjugated molecules?

Options :

1. ✖ Mass spectroscopy
2. ✖ Infrared spectroscopy
3. ✖ NMR spectroscopy
4. ✔ Ultraviolet spectroscopy

Q.5 Question id : 8517 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

A Polymer sample contains 30 percent molecules of molecular mass 20,000; 40 percent molecules of molecular mass 30,000 and the rest molecules of molecular mass 60,000. The value of PDI of this polymer is:

Options :

1. ✖ 2.106
2. ✔ 1.204
3. ✖ 1.240
4. ✖ 0.837

Q.6 Question id : 8518 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Which of the following is correct for Ziegler- Natter polymers ?

Options :

1. ✖ They are linear with 70 percent branching
2. ✔ They are linear and have practically no chain branching
3. ✖ They are linear with 30 percent branching
4. ✖ They are stereo-chemically uncontrollable

Q.7 Question id : 8519 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Cationic polymerization is fastest in which of the following?

Options :

1. ✔ 2 – methylpropene
2. ✖ 2 – Chloroprene
3. ✖ Butene – 1
4. ✖ Butene – 2

Q.8 Question id : 8520 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Which of the following ion is not aromatic in character?

Options :

1. ✖ Cyclopentadienyl anion
2. ✖ Cycloheptatrienyl cation
3. ✔ Cyclopentadienyl cation
4. ✖ Cyclopropenyl cation

Q.9 Question id : 8521 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Which of the following is correct for annulenes?

Options :

1. ✔ 1. They are monocyclic conjugated polyenes containing ten or more carbon atoms in the ring
2. ✖ 2. They are bicyclic conjugated polyenes containing ten or more carbon atoms in the ring

3. They are tricyclic conjugated polyenes containing more than ten carbon atoms in the ring

3. ✖

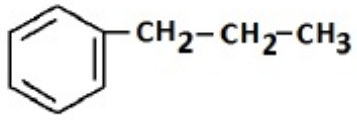
4. They are monocyclic conjugated polyenes containing eight carbon atoms in the ring

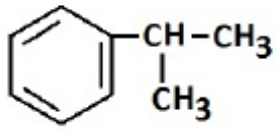
4. ✖

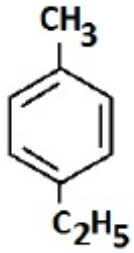
Q.10 Question id : 8522 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

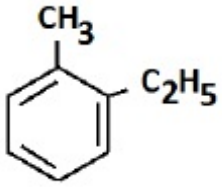
The major product obtained by the reaction of benzene with 1 – Bromopropane in the presence of FeBr3 is:

Options :

1. ✖ 

2. ✔ 

3. ✖ 

4. ✖ 

Q.11 Question id : 8523 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Out of lactic acid, 2–Bromobutane, 2–methylcyclohexanone and methylcyclohexane, achiral molecule is :

Options :

1. ✖ Lactic acid

2. ✔ Methylcyclohexane

3. ✖ 2–Bromobutane

4. ✖ 2–Methylcyclohexanone

Q.12 Question id : 8524 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

. Which of the following does not hold good for $[\alpha]_D$, the specific rotation of a compound?

Options :

1. ✖ Sample pathlength (ℓ) is 1 decimeter
2. ✖ Sample concentration C is 1g/mL
3. ✔ Sample pathlength (ℓ) is 1 meter
4. ✖ Light of 589 nanometer (nm) wavelengths is used

Q.13 Question id : 8525 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Which of the following is correct for mesotartaric acid?

Options :

1. ✖ It is monohydroxydicarboxylic acid
2. ✖ It is optically inactive due to external compensation
3. ✖ It is optically active due to internal compensation
4. ✔ It is optical inactive due to internal compensation

Q.14 Question id : 8526 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Correct decreasing order of stability of different conformations of cyclohexane is :

Options :

1. ✖ Twist form > Boat form > Chair form > Half chair form
2. ✔ Chair form > Twist form > Boat form > Half chair form
3. ✖ Half chair form > Boat form > Twist form > Chair form
4. ✖ Chair form > Half chair form > Boat form > Twist form

Q.15 Question id : 8527 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Which of the following is not correct for pericyclic reactions?

Options :

1. ✖ They are single stage concerted reactions
2. ✖ They are intramolecular electrocyclic reactions

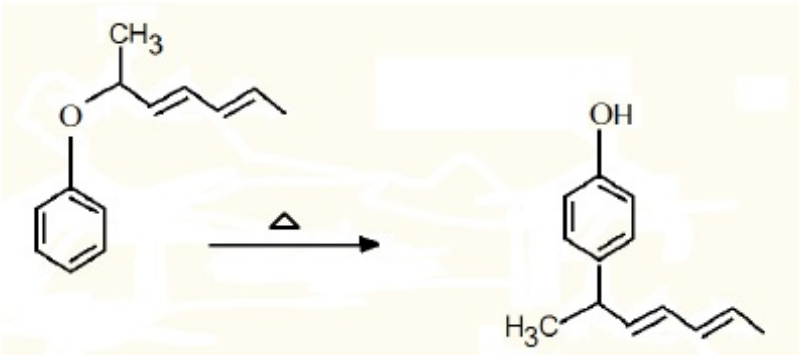
3. ✖ They are cycloaddition reactions

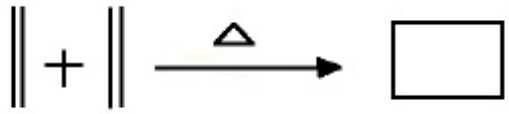
4. ✔ They are shown only by alicyclic compounds

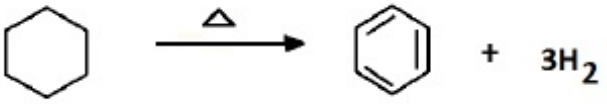
Q.16 Question id : 8528 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

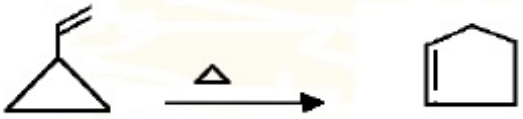
Which of the following is the example of sigmatropic change?

Options :

1. ✔ 

2. ✖ 

3. ✖ 

4. ✖ 

Q.17 Question id : 8529 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Cope rearrangement is exhibited by:

Options :

1. ✖ Conjugated dienes

2. ✔ Biallylic hydrocarbons

3. ✖ Conjugated trienes

4. ✖ r ɣ–alkylallylaryl ethers

Q.18 Question id : 8530 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Thiophene is synthesized industrially by cyclisation of:

Options :

1. ✖ Butene – 1

2. ✖ Butene – 2

3. ✔ 1,3 – Butadiene

4. ✖ 1,4 – Pentadiene

Q.19 **Question id : 8531 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)**

The usual reactivity order of pyrrole, furan, and thiophene towards electrophillic substitution reactions is :

Options :

1. ✖ Thiophene > Pyrrole > Furan

2. ✔ Furan > Pyrrole > Thiophene

3. ✖ Pyrrole > Furan > Thiophene

4. ✖ Thiophene > Furan > Pyrrole

Q.20 **Question id : 8532 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)**

Which of the following reaction is not involved during skraup synthesis of quinoline?

Options :

1. ✖ Dehydration

2. ✖ 1:4 addition

3. ✔ Hydrogenation

4. ✖ Cyclisation

Q.21 **Question id : 8533 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)**

Benzopyrrole has:

Options :

1. ✔ 4 π and 17 σ bonds

2. ✖ 4 π and 16 σ bonds

3. ✖ 3 π and 17 σ bonds

4. ✖ 4 π and 18 σ bonds

Q.22 **Question id : 8534 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)**

Which of the following is correct for killiani- Fischer Synthesis?

Options :

1. ✔ Reaction of aldose with HCN followed by hydrolysis and reduction
2. ✖ Reaction of aldose with KCN followed by hydrolysis and oxidation
3. ✖ Reaction of aldose with HCN followed by hydrolysis and oxidation
4. ✖ Reaction of ketose with HCN followed by hydrolysis and reduction

Q.23 Question id : 8535 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Which of the following hormone contains a benzene like aromatic ring?

Options :

1. ✖ Testosterone
2. ✖ Androsterone
3. ✖ Androstenedione
4. ✔ Estrone

Q.24 Question id : 8536 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Main components of plant essential oils are:

Options :

1. ✖ Carbohydrates
2. ✔ Terpenes
3. ✖ Steroids
4. ✖ Proteins

Q.25 Question id : 8537 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Which of the following α - amino acid has the lowest value of isoelectric points ?

Options :

1. ✖ Glutamic acid
2. ✖ Glycine
3. ✔ Aspartic acid

4. ✖ Cysteine

Q.26 Question id : 8538 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Which of the following is not present in DNA?

QUESTION CANCELLED

Q.27 Question id : 8539 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Which of the following is not correct for enzymes?

Options :

1. ✖ Enzymes are colloidal in nature
2. ✔ They usually contain C, H, N and halogens
3. ✖ The optimum temperature for enzymatic action is between 20 - 40°C
4. ✖ The optimum pH for most of the enzyme actions is about 7

Q.28 Question id : 8540 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Which of the following does not belong to the category of proteins ?

Options :

1. ✖ Hair
2. ✖ Skin
3. ✖ Pulses
4. ✔ Sodium palmitate

Q.29 Question id : 8541 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Efficiency of a photochemical reaction is generally measured in terms of:

Options :

1. ✖ Primary quantum yield

2. ✖ Secondary quantum yield

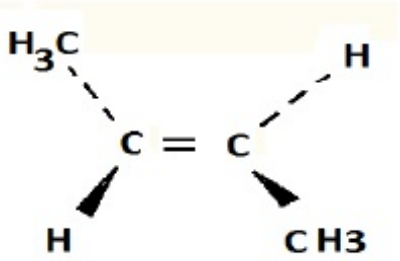
3. ✖ Tertiary quantum yield

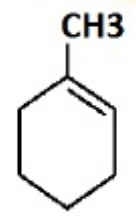
4. ✔ Overall quantum yield

Q.30 **Question id : 8542 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)**

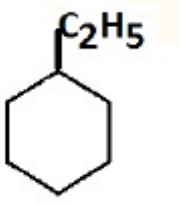
Cis–trans isomerization is not shown by which of the following compound?

Options :

1. ✖ 

2. ✖ 

3. ✖

4. ✔ 

Q.31 **Question id : 8543 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)**

The product obtained by reaction of benzene with excess of chlorine in the presence of sunlight is:

Options :

1. ✔ BHC

2. ✖ C_6H_5Cl

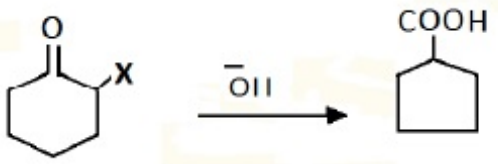
3. ✖ 1; 4 – Dichlorbenzene

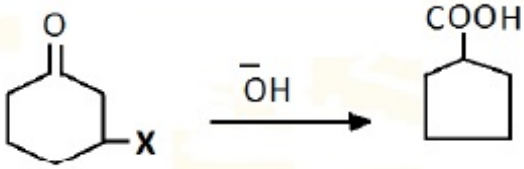
4. ✖ Hexachlorobenzene

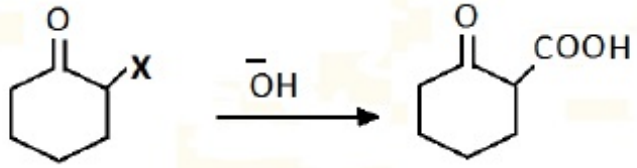
Q.32 **Question id : 8544 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)**

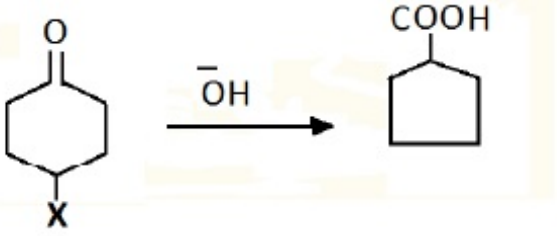
Which of the following is correct for Favorski reaction?

Options :

1. ✔ 

2. ✖ 

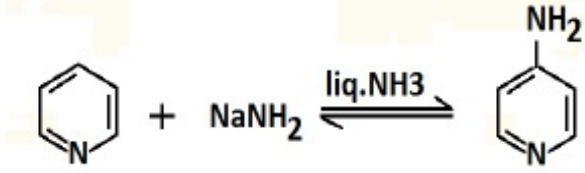
3. ✖ 

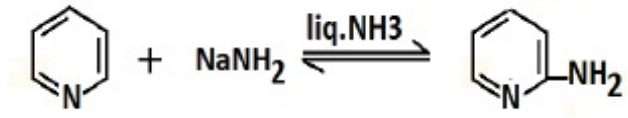
4. ✖ 

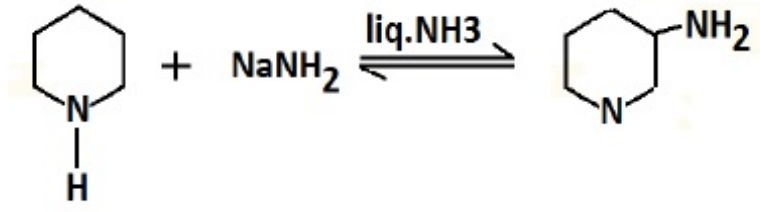
Q.33 Question id : 8545 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

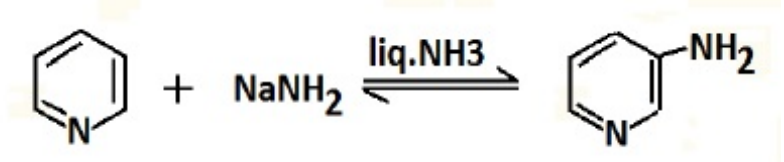
Which of the following is the correct representation of chichibabin reaction?

Options :

1. ✖ 

2. ✔ 

3. ✖ 

4. ✖ 

Q.34 Question id : 8546 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

. The intermediate ion formed during Mannich reaction is:

Options :

1. ✖ Carbanion
2. ✖ Aminoocarbanion
3. ✔ Aminocarbocation
4. ✖ Anilinium ion

Q.35 Question id : 8547 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Which of the following is the correct decreasing order of stability of carbocations?

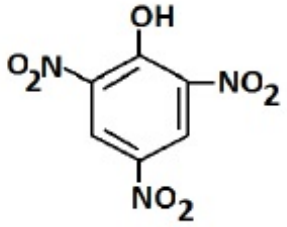
Options :

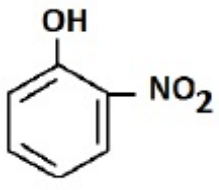
1. ✖ $(CH_3)_3\overset{+}{C} > CH_3CH_2CH_2\overset{+}{C}H_2 > (CH_3)_2\overset{+}{C}H$
2. ✖ $(CH_3)_2\overset{+}{C}H > CH_3CH_2CH_2\overset{+}{C}H_2 > (CH_3)_3\overset{+}{C}$
3. ✖ $(CH_3)_2\overset{+}{C}H > (CH_3)_3\overset{+}{C} > CH_3CH_2CH_2\overset{+}{C}H_2$
4. ✔ . $(CH_3)_3\overset{+}{C} > (CH_3)_2\overset{+}{C}H > CH_3CH_2CH_2\overset{+}{C}H_2$

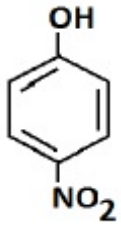
Q.36 Question id : 8548 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

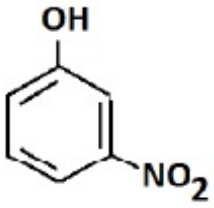
Which of the following is the strongest acid?

Options :

1. ✔ 

2. ✖ 

3. ✖ 

4. ✖ 

Q.37 Question id : 8549 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Which of the following is correct decreasing order of pk_b values for different acids?

QUESTION CANCELLED

Q.38 Question id : 8550 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

pH of 10^{-8} M HCl Solution is :

Options :

1. ✖ 8

2. ✖ 6

3. ✖ 7.6

4. ✔ between 6 and 7

Q.39 Question id : 8551 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Which of the following is correct decreasing bond angle order?

Options :

1. ✖ $NH_3 > H_2O > CCl_4 > BeF_2$

2. ✖ $CCl_4 > H_2O > NH_3 > BeF_2$

3. ✔ $BeF_2 > CCl_4 > NH_3 > H_2O$

4. ✖ $H_2O > NH_3 > CCl_4 > BeF_2$

Q.40 Question id : 8552 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Which of the following has sp^3 hybridization?

Options :

1. ✔ NH_3

2. ✖ $CH_2 = CH - CH = CH_2$

3. ✖ BeF_2

4. ✖ C_6H_6

Q.41 Question id : 8553 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Which of the following equation is used for calculation of frequency (ν) associated with the absorption and emission of the photon?

Options :

1. ✖ $\nu = R_H \times h\left[\frac{1}{n_1^2} - \frac{1}{n_f^2}\right]$

2. ✖ $\nu = \frac{R_H}{h}\left[\frac{1}{n_f^2} - \frac{1}{n_1^2}\right]$

3. ✔ $\nu = \frac{R_H}{h}\left[\frac{1}{n_1^2} - \frac{1}{n_f^2}\right]$

4. ✖ $\nu = R_H \times h\left[\frac{1}{n_f^2} - \frac{1}{n_1^2}\right]$

Q.42 Question id : 8554 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

The orbital with quantum number n = 5 and l = 3 is:

Options :

1. ✖ 5s

2. ✖ 5p

3. ✖ 5d

4. ✔ 5f

Q.43 Question id : 8555 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Correct order of orbitals with increasing energy is :

Options :

1. ✖ $3s < 3p < 4s < 3d$
2. ✖ $4s < 3d < 3p < 3s$
3. ✔ $3s < 3p < 3d < 4d$
4. ✖ $3d < 3p < 3s < 4s$

Q.44 Question id : 8556 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Heisenberg uncertainty principle is represented by:

Options :

1. ✖ $\Delta p\ \Delta x = \frac{1}{8}\left(\frac{h}{2\pi}\right)$
2. ✖ $\Delta p\ \Delta x = \frac{1}{4}\left(\frac{2\pi}{h}\right)$
3. ✖ $\Delta p\ \Delta x \geq \frac{1}{4}\left(\frac{h}{2\pi}\right)$
4. ✔ $\Delta p\ \Delta x \geq \frac{1}{2}\left(\frac{h}{2\pi}\right)$

Q.45 Question id : 8557 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Which of the assumption is wrong for Born-oppenheimer approximation?

Options :

1. ✔ Molecular nuclei move fast
2. ✖ Molecular nuclei move slowly
3. ✖ Molecular nuclei may be treated as stationery
4. ✖ Electrons move in their field

Q.46 Question id : 8558 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Which of the following is wrong for time independent perturbation theory?

Options :

1. ✖ All the terms in the denominator are negative
2. ✖ All the terms in the numerator are positive
3. ✖ Stronger the perturbation, the greater the lowering of the ground state energy

Weaker the perturbation, the greater the lowering of the ground state energy

4. ✔

Q.47 Question id : 8559 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

The quantum numbers of four electrons named as A, B, C & D are given below

A. $n = 3, \ell = 1, m_l = -1, m_s = +1/2$

B. $n = 3, \ell = 2, m_l = 0, m_s = +1/2$

C. $n = 4, \ell = 1, m_l = 0, m_s = +1/2$

D. $n = 4, \ell = 2, m_l = -2, m_s = -1/2$

The correct order of their increasing energy is:

Options :

1. ✔ A < B < C < D
2. ✖ B < A < C < D
3. ✖ D < C < B < A
4. ✖ C < D < B < A

Q.48 Question id : 8560 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

The electronic configuration of [Ar] $4s^1$ $3d^{10}$ belongs to which element ?

Options :

1. ✖ Chromium
2. ✖ Iron
3. ✔ Copper
4. ✖ Zinc

Q.49 Question id : 8561 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Which of the following is not correct for Raman scattering?

Options :

1. ✖ The molecule returns to a different energy level after interaction with the light
2. ✖ The scattered photon has a longer wavelength than the incident photon
3. ✖ The polarisability of the molecule changes with the vibrations

Raman scattering is not detected by shining light of one wavelength on the sample

4. ✔

Q.50 **Question id : 8562 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)**

The spin quantum numbers m_z are related to the magnetic moment along z- axis, μ_z by which of the following?

Options :

1. ✔ $\mu_z = r \frac{h}{2\pi} m_z$

2. ✖ $\mu_z = h \frac{r}{2\pi} m_z$

3. ✖ $\mu_z = h \frac{r}{4\pi} m_z$

4. ✖ $\mu_z = r \frac{h}{4\pi} m_z$

Q.51 **Question id : 8563 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)**

Which of the following is not correct for thermodynamics?

Options :

1. ✖ Thermodynamics deals with energy transformation

2. ✖ Thermodynamics deals with energy changes of macroscopic systems

3. ✔ Thermodynamics deals with energy changes of microscopic systems

4. ✖ Thermodynamics is not concerned about rate of energy transformations

Q.52 **Question id : 8564 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)**

The enthalpy of combustion of methane, graphite and dihydrogen at 298K are $-890.3\ kJ\ mol^{-1}$, $-393.5kJ\ mol^{-1}$, and $-285.8kJ\ mol^{-1}$ respectively. Enthalpy of formation of CH_4 (g) will be:

Options :

1. ✔ $-74.8\ kJ\ mol^{-1}$

2. ✖ $-52.27kJ\ mol^{-1}$

3. ✖ $+74.8\ kJ\ mol^{-1}$

4. ✖ $+52.48\ kJ\ mol^{-1}$

Q.53 **Question id : 8565 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)**

The correct form of Gibb's Helmholtz equation is :

Options :

1. ✔ $\frac{\partial}{\partial T}\cdot\frac{G}{T}=\frac{-H}{T^2}$

2. ✖ $\frac{\partial}{G}\cdot\frac{T}{\partial T}=+\frac{H}{T^2}$

3. ✖ $\frac{\partial}{\partial T}\cdot\frac{T}{G}=\frac{-H}{T^2}$

4. ✖ $\frac{\partial}{\partial T}\cdot\frac{G}{T}=\frac{-T^2}{H}$

Q.54 Question id : 8566 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

The equilibrium constant for a gas reaction

$4\ NH_{3(g)} + 5\ 0_{2(g)} \rightleftharpoons 4\ NO(g) + 6\ H_2O(g)$ is:

Options :

1. ✖ $\frac{[4\ NH_3][5\ O_2]}{[4\ No][6H_2O]}$

2. ✖ $\frac{[NH_3]^4[O_2]^5}{[No]^4[H_2O]^6}$

3. ✔ $\frac{[No]^4[H_2O]^6}{[NH_3]^4[O_2]^5}$

4. ✖ $\frac{[NH_3]^4[O_2]^5}{[4\ No]\ [6H_2O]}$

Q.55 Question id : 8567 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

27.6 g of N_2O_4 was placed in a one litre reaction vessel at 400K and allowed to attain equilibrium $N_2O_{4(g)} \rightleftharpoons 2NO_2\ (g)$. Using the gas equation Pv = nRT, the value of P is:

Options :

1. ✖ 4.98 bar

2. ✔ 9.96 bar

3. ✖ 2.49 bar

4. ✖ 1.285 bar

Q.56 Question id : 8568 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

The correct form of Debye- Hückel limiting law for very dilute aqueous solution is :

Options :

1. ✔ $\log \gamma_{\pm} = -|z_+ z_-| AI^{1/2}$
2. ✖ $\log \gamma_{\pm} = -|z_+ z_-| AI^{2}$
3. ✖ $\log \gamma_{\pm} = -|z_+ z_-| AI^{1/4}$
4. ✖ $\log \gamma_{\pm} = -|z_+ z_-| AI^{8}$

Q.57 Question id : 8569 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

For a general electrochemical reaction of the type: aA + bB $\xrightarrow{ne^-}$ $cC + dD$, the correct form of Nernst equation is :

Options :

1. ✖ -0.036
2. ✖ -0.092
3. ✔ 0.92
4. ✖ 1.92

Q.58 Question id : 8570 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

The colour intensity of the solution in the equilibrium

$Fe^{3+}(aq) + SCN^{-1}(aq) \rightleftharpoons [Fe(SCN)]^{2+}$ can be increased by:

Options :

1. ✖ adding oxalic acid
2. ✖ adding aq. $HgCl_2$
3. ✔ adding potassium thiocyanate
4. ✖ adding $[Fe(SCN)]^{2+}$

Q.59 Question id : 8571 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

59. For a general electrochemical reaction of the type: aA + bB $\xrightarrow{ne^-}$ $cC + dD$, the correct form of Nernst equation is :

Options :

1. ✔ $E_{cell} = E^{\ominus}_{cell} - \frac{RT}{nF} \ln \frac{[C]^c[D]^d}{[A]^a[B]^b}$

2. ✖ $E_{cell} = E^{\ominus}_{cell} + \frac{RT}{nF} \ln \frac{[C]^c[D]^d}{[A]^a[B]^b}$

3. ✖ $E^{\ominus}_{cell} = E_{cell} - \frac{RT}{nF} \ln \frac{[C]^c[D]^d}{[A]^a[B]^b}$

4. ✖ $E_{cell} = E^{\ominus}_{cell} - \frac{RT}{nF} \ln \frac{[A]^a[B]^b}{[C]^c[D]^d}$

Q.60 Question id : 8572 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

The electricity required in terms of Faraday to produce 20 g of Ca from molten $CaCl_2$ is:

Options :

1. ✖ 4F
2. ✖ 2 F
3. ✖ ½ F
4. ✔ 1 F

Q.61 Question id : 8573 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Which of the following processes does not contain interface?

Options :

1. ✖ Crystallization of potash alum
2. ✖ Rusting of iron
3. ✔ Dissolution of H_2 in O_2
4. ✖ Dissolution of sugar in water

Q.62 Question id : 8574 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Which of the following is the example of sorption?

Options :

1. ✔ Dipping of a chalk stick in blue ink
2. ✖ Exposure of anhydrous $CaCl_2$ to moisture
3. ✖ Exposure of silica gel bottle to moisture
4. ✖ Passing of aqueous solution of raw sugar over a bed of animal charcoal

Q.63 Question id : 8575 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

The substance used in photographic films is :

Options :

1. ✖ KBr
2. ✖ $AlBr_3$
3. ✖ AgCl
4. ✔ AgBr

Q.64 **Question id : 8576 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)**

The technique used for studying the surface state of heterogeneous catalysts is:

Options :

1. ✔ ESCA
2. ✖ NMR
3. ✖ X-ray
4. ✖ u.v.

Q.65 **Question id : 8577 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)**

The value of P in Fermi-Dirac distribution is :

Options :

1. ✔ 0.5
2. ✖ 2.0
3. ✖ 1.0
4. ✖ 0.25

Q.66 **Question id : 8578 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)**

According to Maxwell- Boltzman distribution of velocities, the correct relationship between (u), $(u^2)^{1/2}$ and $(u^3)^{1/3}$ is :

Options :

1. ✖ $u > (u^2)^{1/2} > (u^3)^{1/3}$
2. ✖ $u > (u^3)^{1/3} > (u^2)^{1/2}$
3. ✔ $(u^3)^{1/3} > (u^2)^{1/2} > u$
4. ✖ $(u^2)^{1/2} > u > (u^3)^{1/3}$

Q.67 **Question id : 8579 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)**

Which of the following is the correct form of Gibb's free energy equation?

Options :

1. ✖ $\Delta H = \Delta G - T\Delta S$
2. ✔ $\Delta G = \Delta H - T\Delta S$
3. ✖ $\Delta S = \Delta G - T\Delta H$
4. ✖ $\Delta G = \Delta H + T\Delta S$

Q.68 Question id : 8580 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

For the reaction $Hg(l) + Cl_2(g) \rightarrow HgCl_2(s)$, the correct way of representation of rate of reaction is :

Options :

1. ✖ Rate of reaction $= \frac{\Delta[Hg]}{\Delta t}$
2. ✖ Rate of reaction $= \frac{[\Delta Cl_2]}{\Delta t}$
3. ✔ Rate of reaction $= \frac{\Delta [HgCl_2]}{\Delta t}$
4. ✖ Rate of reaction $= \frac{-\Delta[HgCl_2]}{\Delta t}$

Q.69 Question id : 8581 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

According Michaelis -Menton kinetics, which of the following relationship is correct, where [E], $[E]_0$ and [ES] carry their usual meanings ?

Options :

1. ✖ $[E] = [E]_0 + [ES]$
2. ✔ $[E_0] = [E] + [ES]$
3. ✖ $[ES] = [E] + [E_0]$
4. ✖ $[ES] = [E[- [E_0]$

Q.70 Question id : 8582 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Which of the following statement is correct for flash photolysis?

Options :

1. ✖ It is used to study rusting of iron
2. ✖ It is used to study hydrolysis of ethyl acetate

3. ✔ It is used to study rate of neutralization reactions

4. ✖ It is used to study rate of burning of a candle

Q.71 Question id : 8583 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Which of the following is not the example of amorphous solid?

Options :

1. ✖ PTFE (Polytetrafluoroethylene)

2. ✖ PAN (Polyacrylonitrile)

3. ✖ Polyisoprene

4. ✔ Sodium chloride

Q.72 Question id : 8584 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Which of the following is wrong for Frankel defect?

Options :

1. ✖ Frankel defect is shown by ionic solids

2. ✖ Due to Frankel defect, a vacancy defect is created

3. ✖ Frankel defect is also known as dislocation defect

4. ✔ Density of the solid is changed due to Frankel defect

Q.73 Question id : 8585 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Which of the following is paramagnetic?

Options :

1. ✔ Cr^{3+}

2. ✖ MnO

3. ✖ NaCl

4. ✖ C_6H_6

Q.74 Question id : 8586 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

The daughter element obtained by emission of one alpha and two beta particles from $^{238}_{92}U$ is:

Options :

1. ✖ $^{236}_{92}U$

2. ✔ $^{234}_{92}U$

3. ✖ $^{234}_{91}Pa$

4. ✖ $^{234}_{90}Th$

Q.75 Question id : 8587 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

The neutron proton ratio after emission of two alpha particles from $^{234}_{90}Th$ will be :

Options :

1. ✖ 1.70

2. ✖ 1.56

3. ✖ 1.50

4. ✔ 1.60

Q.76 Question id : 8588 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Number of alpha (α) and Beta (β) particles emitted during radioactive decay of $^{238}_{92}U$ to $^{206}_{82}Pb$ are:

Options :

1. ✖ 10 β and 8 α

2. ✔ 8 α and 6 β

3. ✖ 6 α and 8 β

4. ✖ 4 α and 10 β

Q.77 Question id : 8589 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Which of the following is not correct pair of isomorphous substances?

Options :

1. ✖ $ZnSO_4 . 7H_20$ and $FeS0_4 . 7H_2O$

2. ✖ $K_2SO_4 . (Cr_2SO_4)_3 . 24H_2O$ and $K_2SO_4 . (Al_2SO_4)_3 . 24H_2O$

3. ✖ ½ $[K_2SO_4. (Cr_2SO_4)_3].12H_2O$ and ½ $[K_2SO_4.(Al_2SO_4)_3]. 12H_2O$

4. ✔ $FeSO_4 . 7H_2O$ and $Fe(SO_4)_3 . 7H_2O$

Q.78 Question id : 8590 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Sulphur molecules are best represented as:

Options :

1. ✖ S_2
2. ✖ S_4
3. ✔ S_8
4. ✖ S

Q.79 Question id : 8591 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Out of $MgCl_2$, $CaCl_2$, $SrCl_2$, and $BaCl_2$, which has got the highest tendency for hydrate formation?

Options :

1. ✔ $MgCl_2$
2. ✖ $CaCl_2$
3. ✖ $SrCl_2$
4. ✖ $BaCl_2$

Q.80 Question id : 8592 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Which of the following is dibasic acid?

Options :

1. ✖ H_3PO_4
2. ✔ H_3PO_3
3. ✖ H_3PO_5
4. ✖ $(HPO_3)_n$

Q.81 Question id : 8593 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Out of ClO^-, ClO_2^-, ClO_3^- and ClO_4^- which has tetrahedral shape?

Options :

1. ✖ ClO_2^-
2. ✔ ClO_4^-
3. ✖ ClO^-

4. ✖ ClO_3^-

Q.82 Question id : 8594 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Hydrolysis product of borazole is:

Options :

1. ✖ C_2H_5OH
2. ✖ C_6H_5OH
3. ✖ $C_2H_5NH_2$
4. ✔ H_3BO_3

Q.83 Question id : 8595 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Which of the following is obtained by distilling concentrated Nitric acid over phosphorus Pentoxide?

Options :

1. ✖ N_2O
2. ✖ N_2O_3
3. ✔ N_2O_5
4. ✖ NO_2

Q.84 Question id : 8596 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

The correct order of increasing pK_b value of KOH, NaOH & LiOH is :

Options :

1. ✔ $KOH < NaOH < LiOH$
2. ✖ $KOH > NaOH > LiOH$
3. ✖ $LiOH < NaOH < KOH$
4. ✖ $NaOH < KOH < LiOH$

Q.85 Question id : 8597 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

The decreasing order of pK_a value of $HClO_4$, $HClO_3$ and $HClO_2$ is:

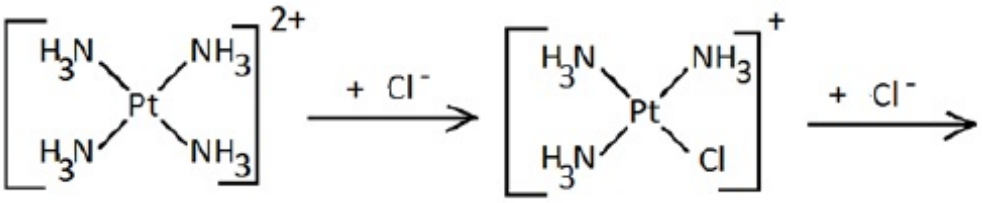
Options :

1. ✖ $HClO_4 > HClO_3 > HClO_2$
2. ✔ $HClO_2 > HClO_3 > HClO_4$
3. ✖ $HClO_3 > HClO_2 > HClO_4$

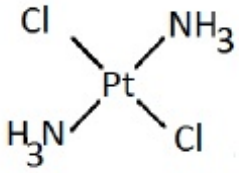
4. ✖ $HClO_4 > HClO_2 > HCl_3$

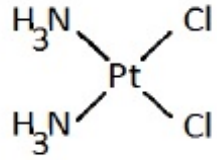
Q.86 Question id : 8598 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

The exclusive product formed in the reaction given below is :



Options :

1. ✔


2. ✖


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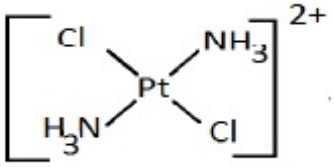

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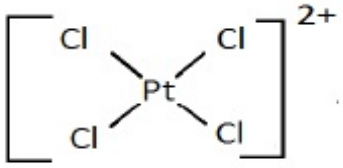

Q.87 Question id : 8599 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

The highest magnetic moment (μ) is of:

Options :

1. ✖ V^{4+}
2. ✖ V^{3+}
3. ✖ Cr^{3+}
4. ✔ Mn^{2+}

Q.88 Question id : 8600 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

The maximum number of oxidation states are shown by:

Options :

1. ✖ Cu
2. ✔ Mn
3. ✖ Cr
4. ✖ Ni

Q.89 **Question id : 8601 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)**

Which of the following ions is coloured?

Options :

1. ✔ V^{4}
2. ✖ Sc^{3+}
3. ✖ Ti^{4+}
4. ✖ Zn^{2+}

Q.90 **Question id : 8602 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)**

Which of the following has tetrahedral geometry?

Options :

1. ✖ $PtCl_4^{2-}$
2. ✖ CoF_6^{3-}
3. ✔ $NiCl_4^{2-}$
4. ✖ $Co(NH_3)_6^{3+}$

Q.91 **Question id : 8603 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)**

The paramagnetic moment (μ) for lanthanide complexes can be calculated by which of the following, where symbols carry their usual meanings?

Options :

1. ✔ $\mu = g[J(J+1)]^{\frac{1}{2}}$
2. ✖ $\mu = g[J(J-1)]^{\frac{1}{2}}$
3. ✖ $\mu = g[J(J+1)]^{2}$
4. ✖ $\mu = g[J(J-1)]^{2}$

Q.92 **Question id : 8604 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)**

Atomic number of lanthanoid element with electronic configuration $[Xe]4f^{11}6s^{2}$ is :

Options :

1. ✖ 49
2. ✔ 67
3. ✖ 69
4. ✖ 99

Q.93 Question id : 8605 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

IUPAC name of the compound obtained by reaction of $C_2H_5 - MgBr$ and propanone in dry ether solvent followed by hydrolysis is :

Options :

1. ✖ 2 – Butanol
2. ✖ 2 – Ethyl – 2- Propanol
3. ✔ 2 – Methyl – 2 – butanol
4. ✖ 3 – Methyl – 2 – pentanol

Q.94 Question id : 8606 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Which of the following does not exist as a dimer?

Options :

1. ✔ SO_2
2. ✖ $AlCl_3$
3. ✖ P_2O_5
4. ✖ $Mn(CO)_5$

Q.95 Question id : 8607 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Which of the following is not associated with nitrogen fixation?

Options :

1. ✖ Vivo nitrogen fixation
2. ✖ Vitro nitrogen fixation
3. ✔ Carbo nitrogen fixation
4. ✖ Nitrogenase enzyme

Q.96 Question id : 8608 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

The order of stability of complexes of porphyrins with +2 metal ions is:

Options :

1. ✔ $Ni^{2+} > Cu^{2+} > Fe^{2+} > Zn^{2+}$
2. ✖ $Zn^{2+} > Fe^{2+} > Cu^{2+} > Ni^{2+}$
3. ✖ $Cu^{2+} > Fe^{2+} > Zn^{2+} > Ni^{2+}$
4. ✖ $Fe^{2+} > Zn^{2+} > Cu^{2+} > Ni^{2+}$

Q.97 Question id : 8609 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Which of the following is not correct for cytochromes?

Options :

1. ✖ They are heme proteins
2. ✖ They act as electron carriers
3. ✖ They are enzymes
4. ✔ They can convert n - hexane into benzene

Q.98 Question id : 8610 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Which of the following is not associated with solvent extraction?

Options :

1. ✖ Continuous boiling with solvent
2. ✖ Used for extraction of essential oils from plants
3. ✔ Used for petroleum refining
4. ✖ Slow but continuous process

Q.99 Question id : 8611 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)

Which of the following is not used as a dehydrating reagent?

Options :

1. ✖ $KHSO_4$
2. ✖ Anhyd $ZnCl_2$
3. ✖ Conc. H_2SO_4
4. ✔ Liquid bromine

Q.100 **Question id : 8612 Question Type : MCQ (Correct + 2.0 , Wrong - 0.0)**

The reagent used in Clemmenson reduction is :

Options :

1. ✖ $LiAlH_4$
2. ✖ $Na/Hg + C_2H_5OH$
3. ✔ $Zn/Hg + Conc.\ HCl$
4. ✖ $(Me_2CHO)_3Al$